

Introduction



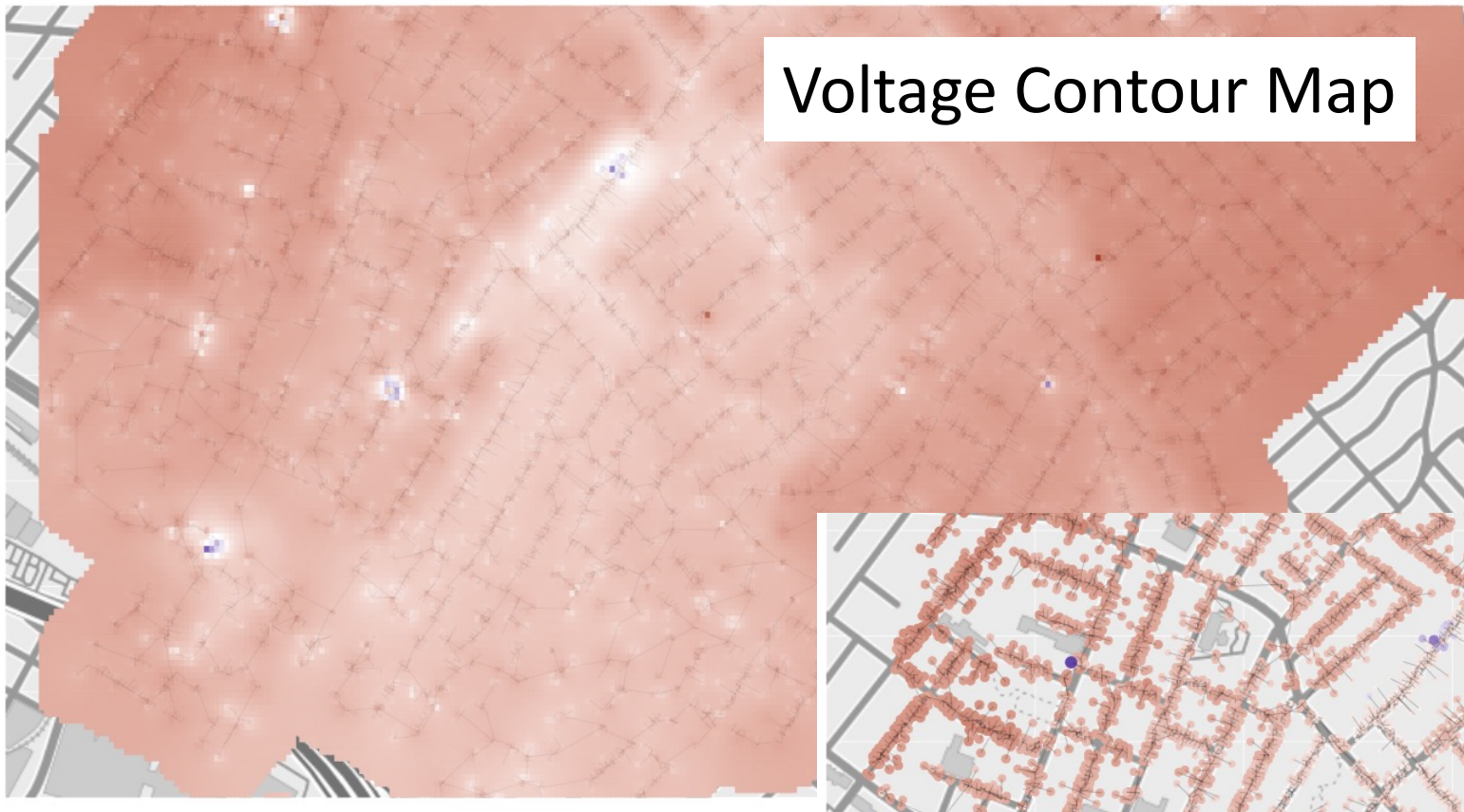
- Electricity grid models are growing **larger** and more **complex** and need good visualizations
- Voltages on a grid need to **stay within 5%** of the expected value, 1.00 per unit voltage
- Integration of renewables and new loads like EV charging is **straining existing grid infrastructure**
- Grid planners need to know what **system upgrades** will need to be made to prepare for higher EV and solar adoption
- We can model an **electrical network** of thousands of nodes and edges

Research Questions: How do we visualize the voltage on an electrical grid to figure out where there might be problems? How can we aggregate the voltages in ways that are accurate to the underlying data?

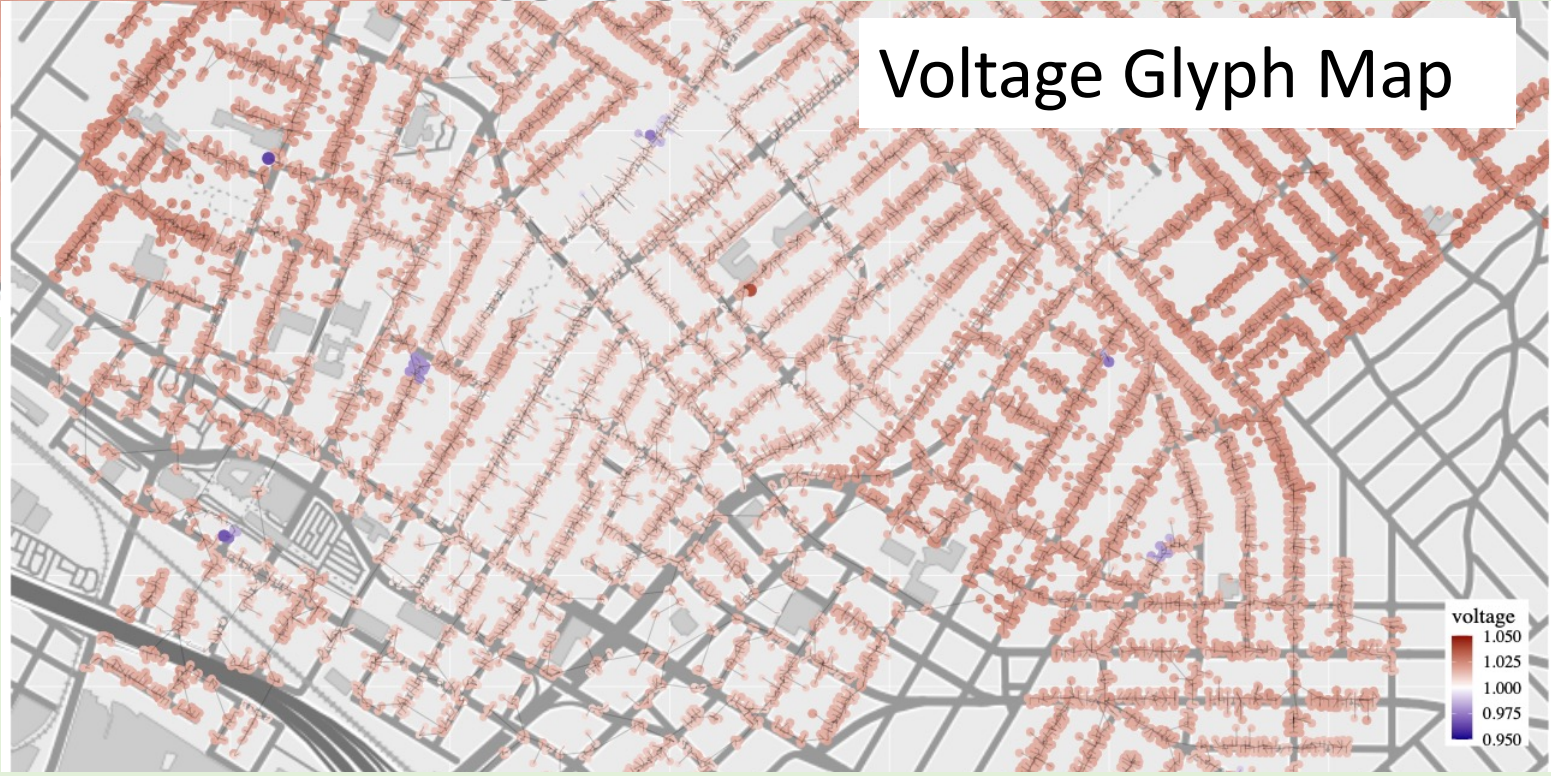
Related Work



Dataset: SFO OpenDSS Model



Voltage Contour Map



Voltage Glyph Map

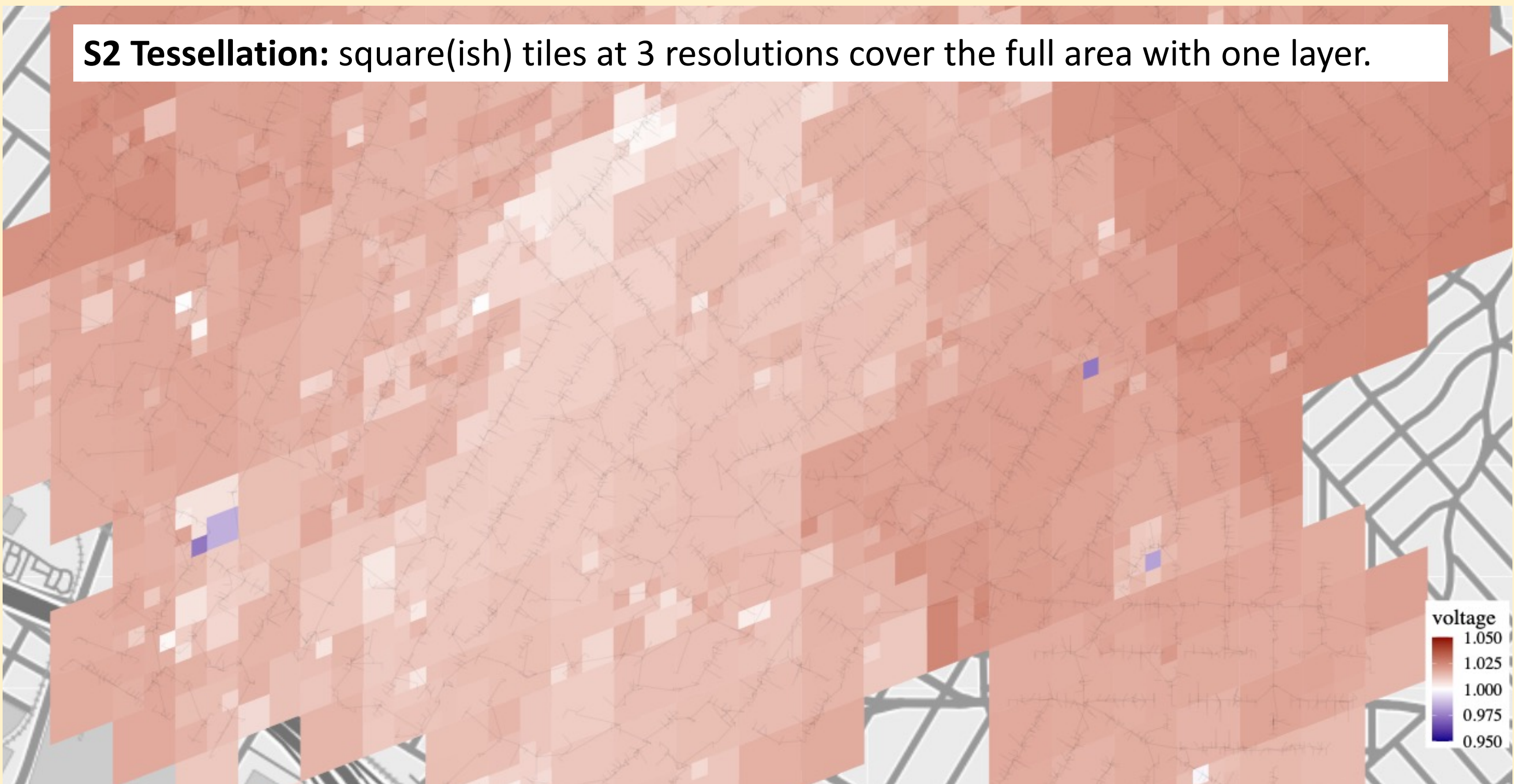
Dataset: the data includes 24,000+ electrical junctions (buses) from a neighborhood in Oakland.

Contours: standard practice is to visualize the state of the grid with a continuous heatmap of voltage.

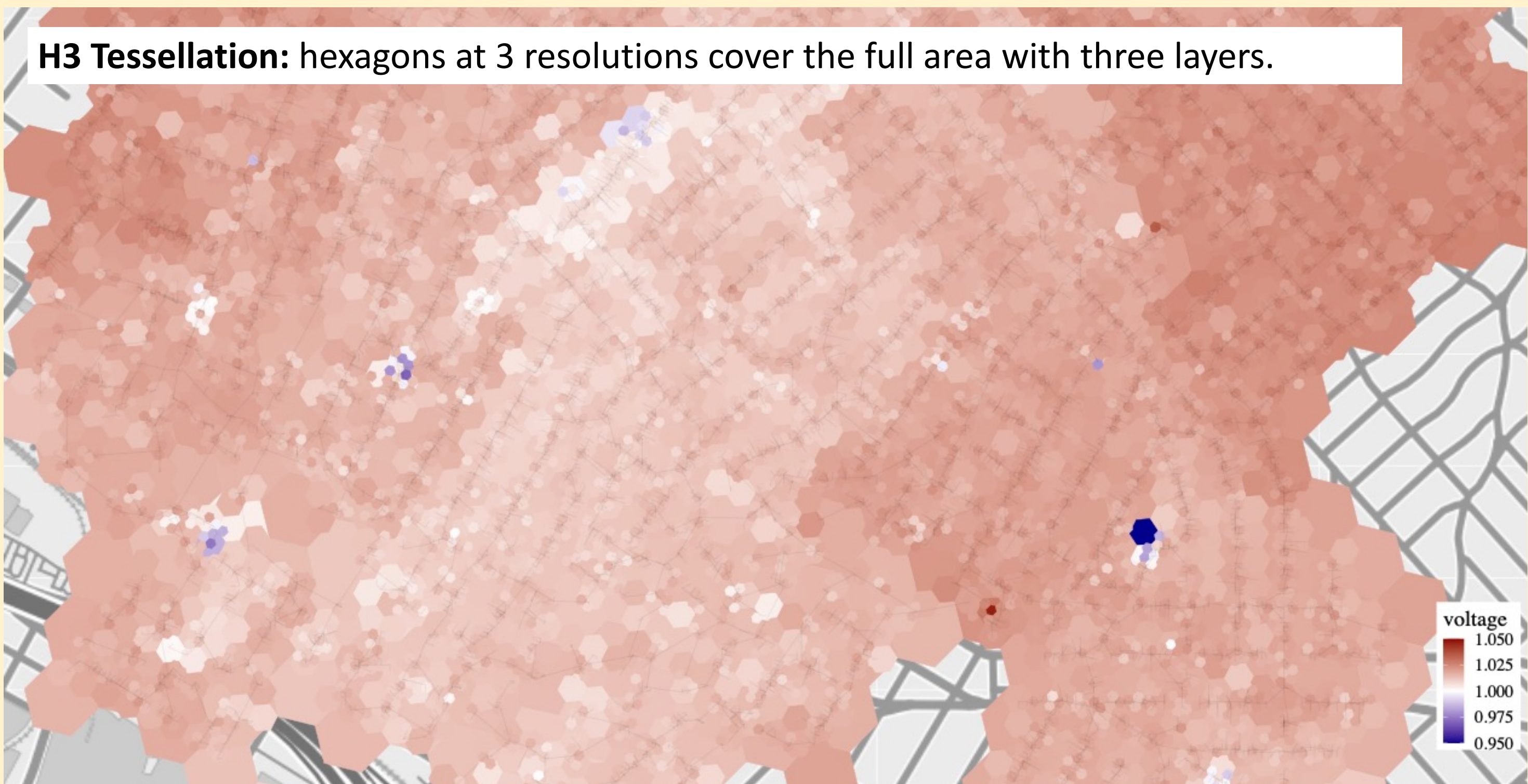
Glyphs: Gruchalla et al. showed last year that using glyphs can help users identify outliers better than contour maps. But can we do better?

Visualizations

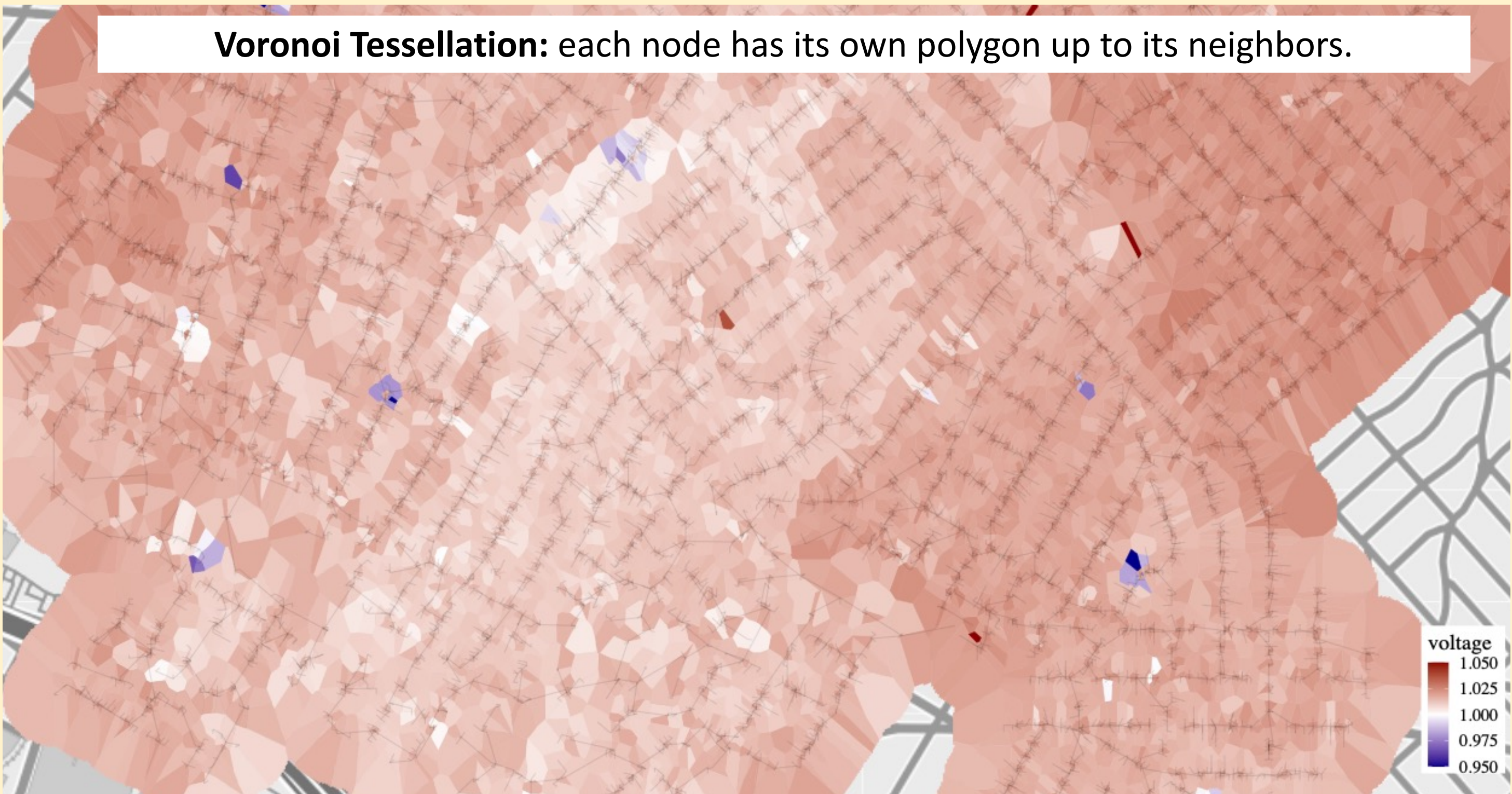
We re-aggregate the data with different tile shapes such as squares (1), hexagons (2), Voronoi polygons (3), and by changing the aggregation algorithm (4) to generate the four visualizations below.



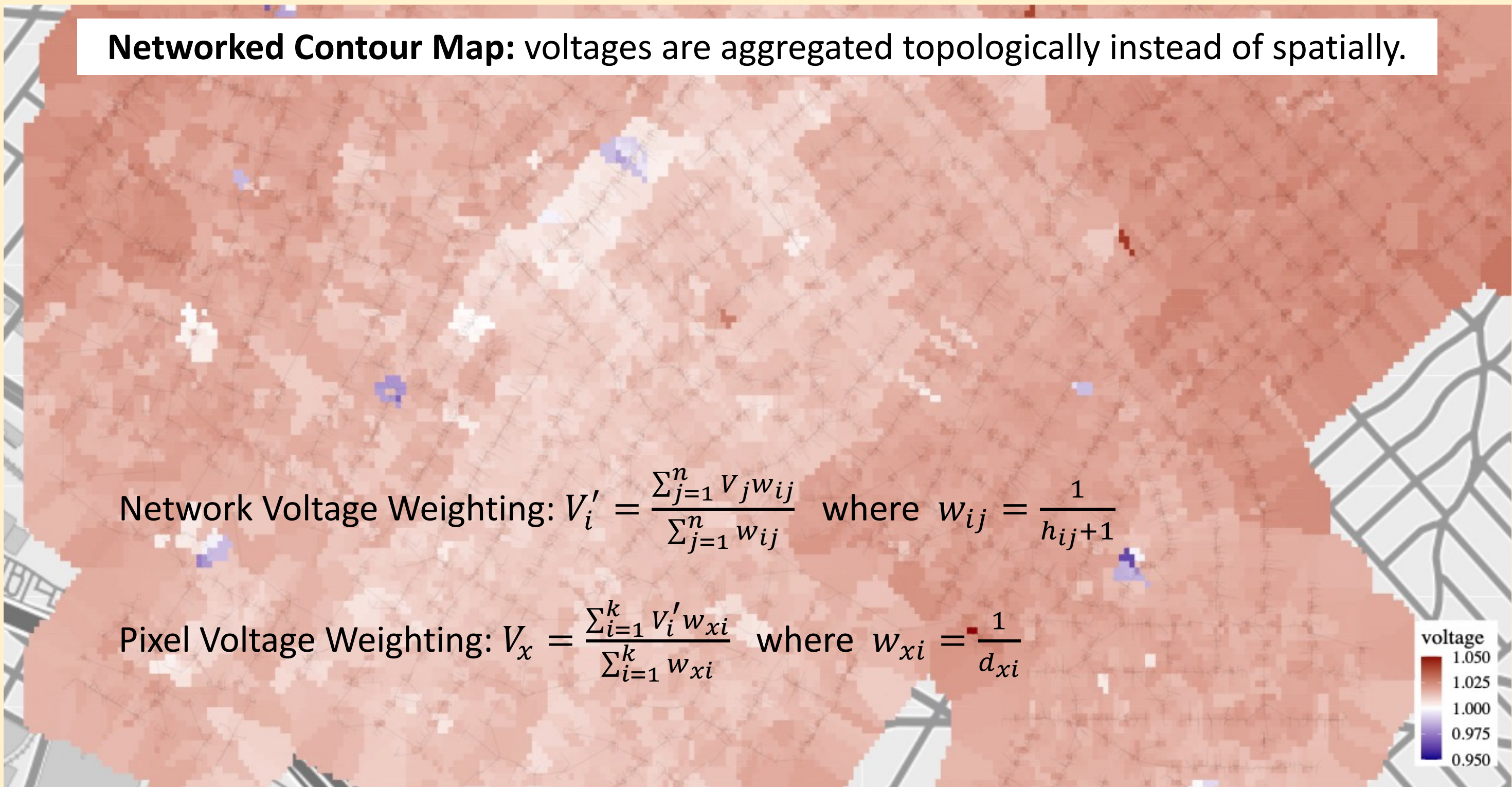
S2 Tessellation: square(ish) tiles at 3 resolutions cover the full area with one layer.



H3 Tessellation: hexagons at 3 resolutions cover the full area with three layers.



Voronoi Tessellation: each node has its own polygon up to its neighbors.

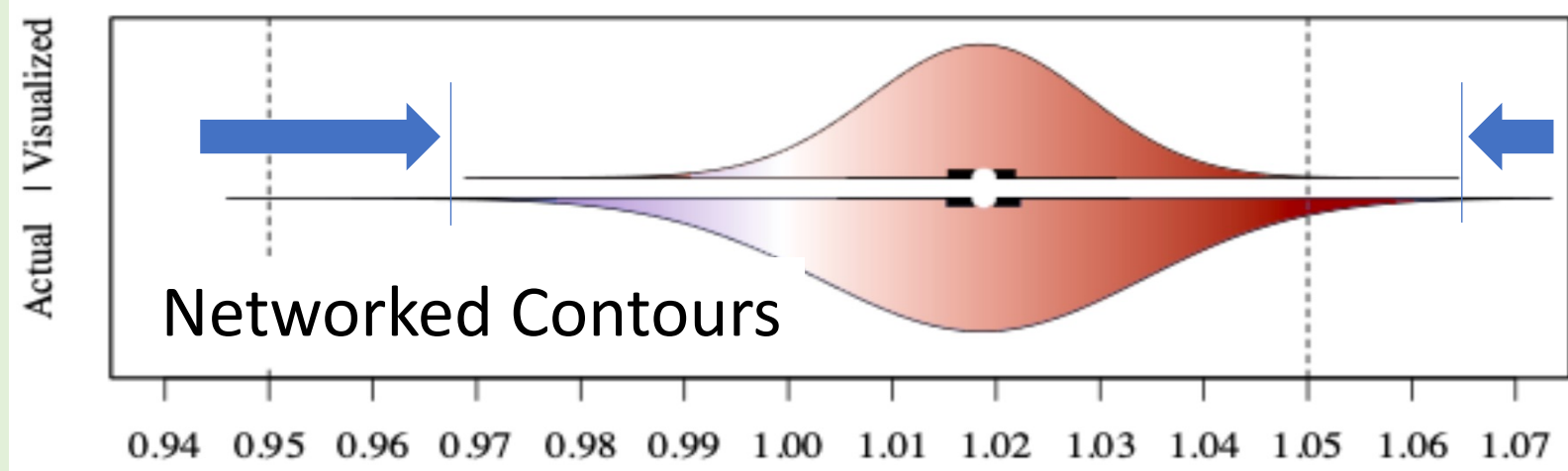
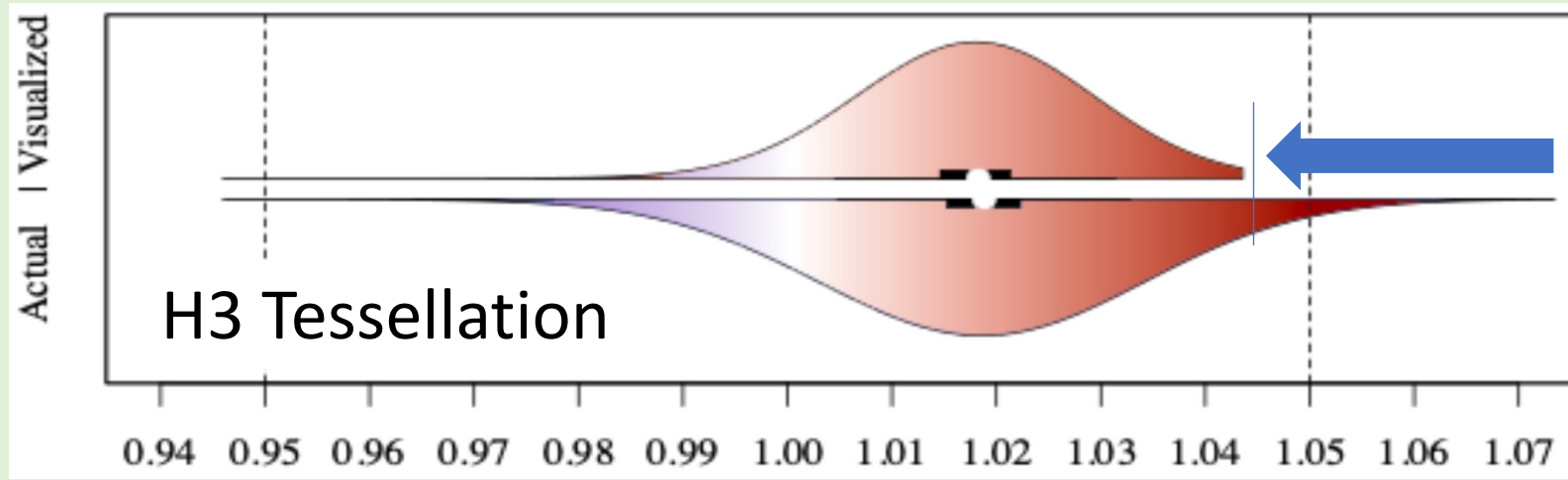
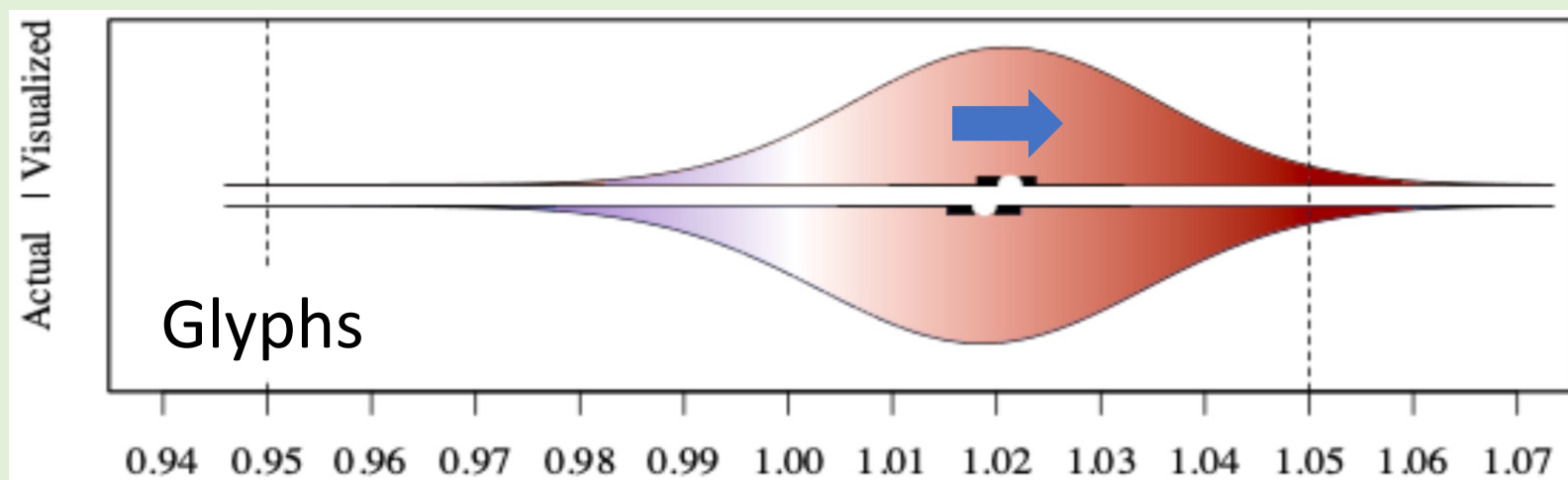
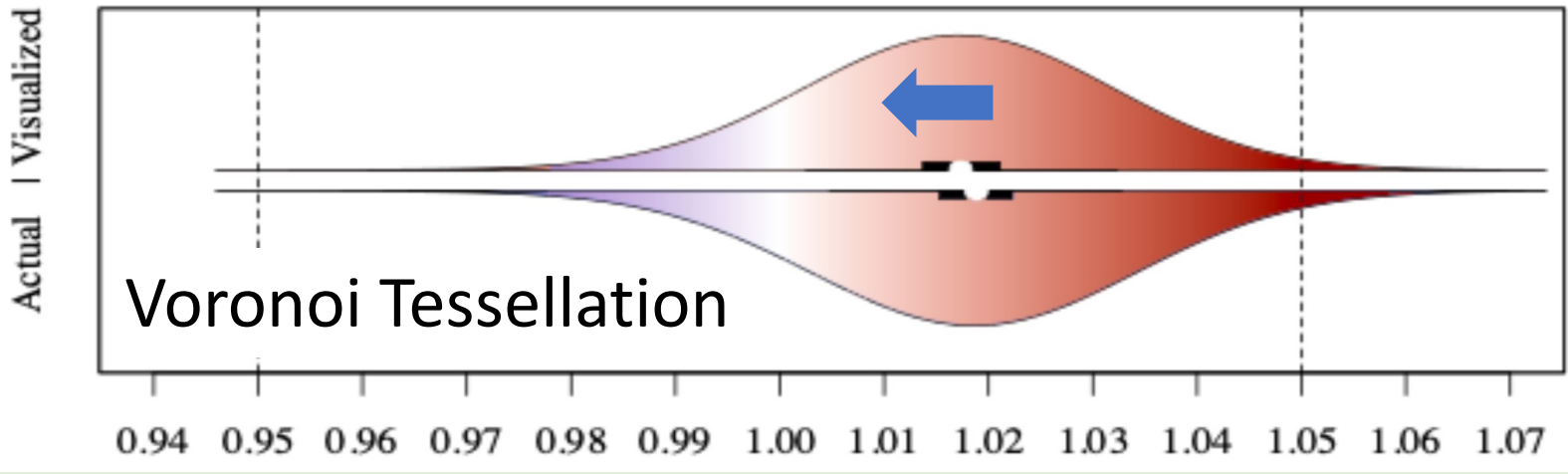
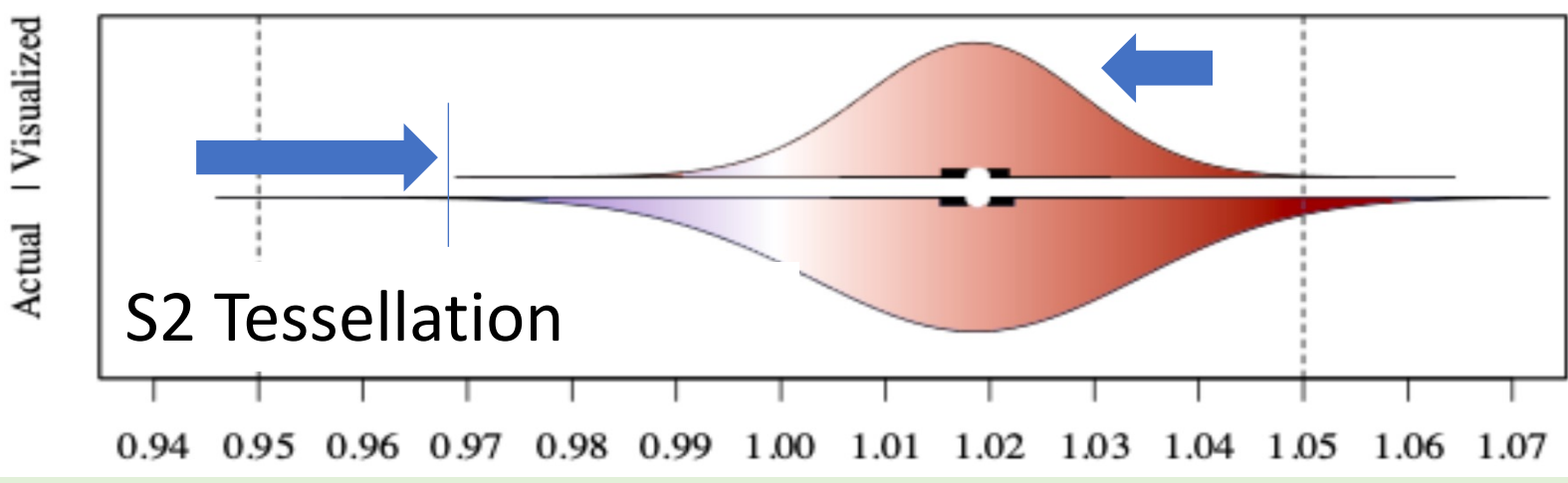
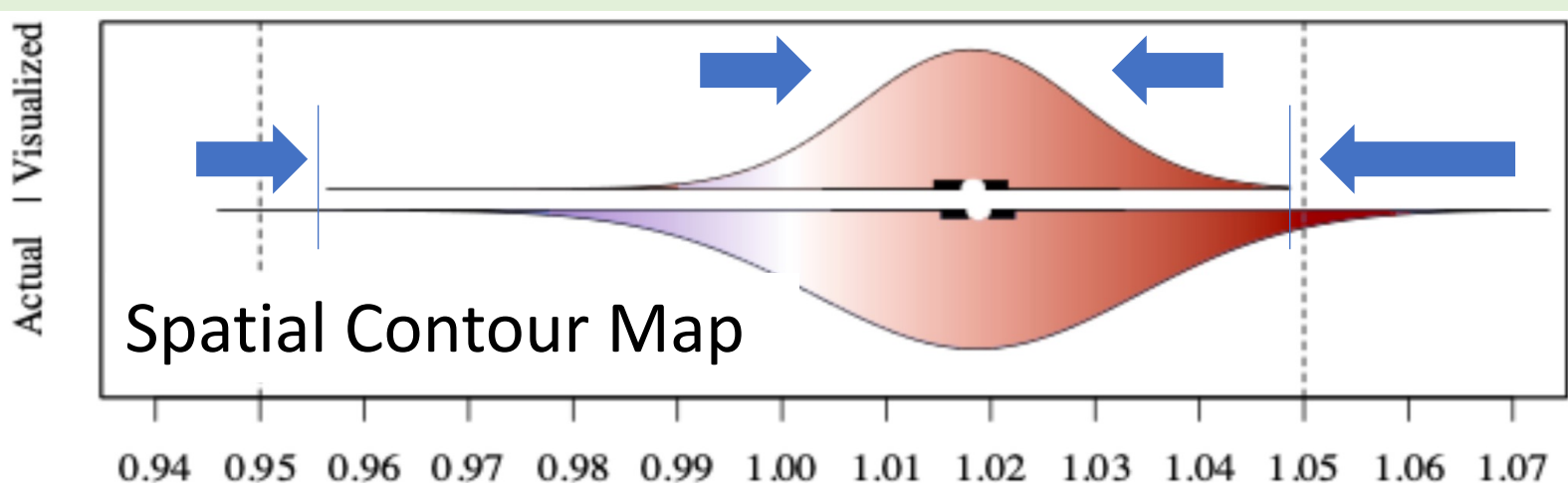


Networked Contour Map: voltages are aggregated topologically instead of spatially.

Network Voltage Weighting: $V'_i = \frac{\sum_{j=1}^n V_j w_{ij}}{\sum_{j=1}^n w_{ij}}$ where $w_{ij} = \frac{1}{h_{ij}+1}$

Pixel Voltage Weighting: $V_x = \frac{\sum_{i=1}^k V'_i w_{xi}}{\sum_{i=1}^k w_{xi}}$ where $w_{xi} = \frac{1}{d_{xi}}$

Evaluation



We compare the distribution of values visualized (top half) with the distribution of actual voltage values (bottom). Here, we can see if the distribution has had the mean shifted, the variance reduced, or the tails cut off. The more vertically symmetrical these diagrams are, the more accurate they are.

Conclusions

Metric	C	G	V	H3	S2	N
Accurate distribution	✓	x	✓	✓	✓	✓
Preserves areas of high variance	x	✓	✓	x	x	✓
Preserves tails / outliers	x	✓	✓	x	x	x
Computationally efficient	x	✓	✓	x	x	x

- We found that the top performing visualization was the **Voronoi Tessellation**.
- We suspect that this is because we have more pixels than network nodes, so each one can be assigned a polygon.
- The **Networked Contour** outperformed the spatial contour. We think this holds promise for larger datasets where you have more network nodes than pixels.
- The **H3** and **S2 Tessellations** did similar well to the Spatial Contour Map.

What do you think? Which visualization do you like the best? What should we test next?